

practical comparison. “Think of AI as a car engine. That is deep learning. Put it into a car and you have generative AI; it drives, but only when instructed. Now consider a self-driving car. It plans, acts, reacts, reflects, and adapts. That is autonomous AI. The last two—reflecting and adapting—are what make it an autonomous agent.”

The panel agrees that this shift was not the result of a single breakthrough, but of steady evolution. Advances include expanded context windows in large language models (LLMs), improved memory management to retrieve past prompts, function calling for external data access, and multi-agent frameworks that function like coordinated teams of specialists.

Prompt engineering itself illustrates this transition. The efficiency of agents has overtaken prompt engineering as the most sought-after capability. But it is more of an evolution than a mere replacement. Clear instructions still matter, notes Sunil.

Sonia Taneja, Senior Engineering Manager at Nutanix, highlights the tangible impact of autonomous agents across industries. She cites autonomous vehicles that make on-the-fly decisions, even in Bengaluru traffic! In manufacturing, chip design cycles that once took years can now be completed in weeks. In financial operations, organisations such as JP Morgan have reported agents saving hundreds of thousands of human work hours annually. Healthcare applications are also advancing, with radiology systems automatically retrieving prior records, analysing scans, and flagging anomalies that clinicians might miss.

Most users today interact with AI through text-based interfaces. But what happens when agents meet voice?

Dr Devasia Kurian, Founder and Managing Director at *astTECS, describes everyday office scenarios. “Today, I call an administrative assistant to book a ticket to London. In the near future, an agent AI could handle the task

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—Gaurav Gupta, Principal Engineer, Unisys

end to end: checking flights, layovers, and even calling a travel agent. Or consider scheduling a meeting with ten people. The agent would check room availability, book time slots, arrange refreshments, test the projector, and return for approvals. It works like an assistant: some decisions are taken independently, others are confirmed.”

Srivathsa NS, Senior Engineering Director at Unisys, emphasises that these are not theoretical use cases. They are already under active development. When multiple agents operate together, federations and orchestration become critical to ensure coordination and control.

This naturally led to agentic RAG (retrieval-augmented generation), which Gupta describes as the next stage of evolution. “RAG addressed training gaps in LLMs using knowledge bases and digital twins. However, those gaps re-emerged as models aged. Agentic RAG ensures knowledge is continuously updated.” He points to features such as ChatGPT’s internet search capability, which blends trained knowledge with real-time data. “Even RPA is evolving, moving from rule-based workflow automation to decision-making agents.”

While the technology is transformative, human involvement remains a central concern. Srivathsa highlights a key limitation of the human-in-the-loop model, where humans validate or guide AI outputs. While necessary, it can be fatiguing and does not scale effectively.

“Imagine reading four jokes repeatedly, some really bad, and rating them,” he says. “Eventually, human-in-the-loop would not scale.”

Mishra cites AlphaGo as a compelling example of self-learning in

action. The AI played millions of games against itself, discovering moves no human could have conceived, including the famous ‘Move 37,’ which proved decisive in the match. “Agents become truly powerful when they start self-learning, improving themselves without human intervention,” he explains, noting that current agent frameworks are evolving in that direction.

Developed by DeepMind, AlphaGo refined strategies beyond human intuition through repeated self-play. Its famed ‘Move 37,’ initially deemed suicidal by experts, ultimately demonstrated how autonomous agents, using reinforcement learning, can uncover novel solutions and outperform human decision-making without constant human guidance.

Readiness, at individual, technological, and regulatory levels, emerged as a central theme of the discussion. Sonia Taneja emphasises infrastructure readiness, particularly for organisations managing massive data volumes. “Agents feed on data. Someone has to store it, keep it secure, and use it to train the systems,” she says. She also points to the high cost of hardware such as Nvidia GPUs, stressing that organisations must invest mindfully in infrastructure, whether in the cloud or on-premises, to enable agent AI effectively.

From a technological perspective, Gaurav Gupta observes that digitally mature companies with exposed APIs and well-structured data can adopt agents more quickly. In contrast, organisations with siloed and complex systems may struggle. He highlights growing frameworks such as AutoGPT, Crew.ai, and LangGraph, along with platforms hosting thousands of agents,